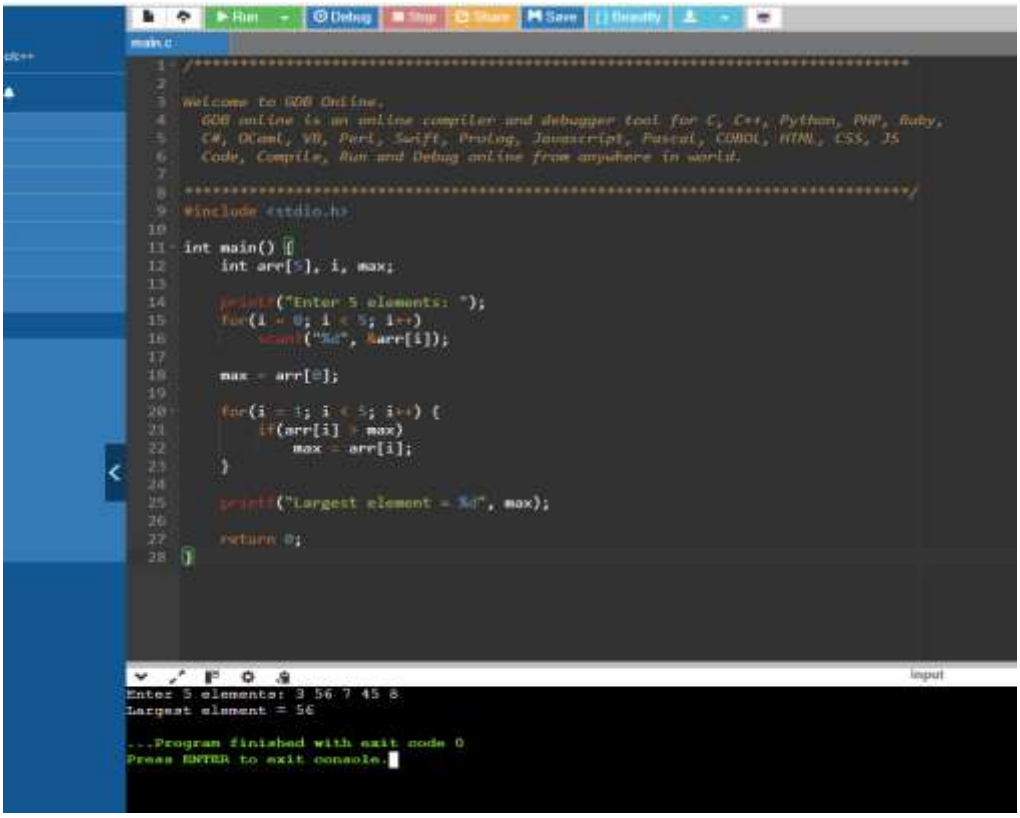
Largest element using array



```
/*******************************************************************************
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*******************************************************************************/
#include <stdio.h>

int main() {
    int arr[5], i, max;

    printf("Enter 5 elements: ");
    for(i = 0; i < 5; i++)
        scanf("%d", &arr[i]);

    max = arr[0];

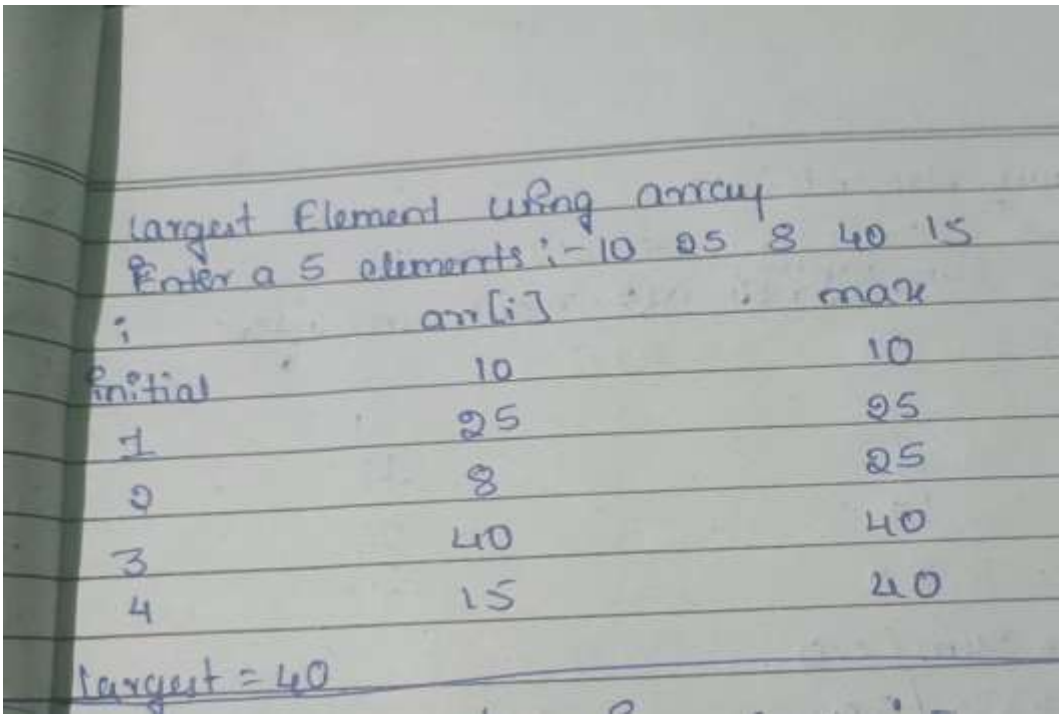
    for(i = 1; i < 5; i++) {
        if(arr[i] > max)
            max = arr[i];
    }

    printf("Largest element = %d", max);

    return 0;
}
```

```
Enter 5 elements: 3 56 7 45 8
Largest element = 56

...Program finished with exit code 0
Press ENTER to exit console.
```



Largest Element using array

Enter a 5 elements :- 10 25 8 40 15

| i | arr[i] | max |
|---|---|---|
| Initial | 10 | 10 |
| 1 | 25 | 25 |
| 2 | 8 | 25 |
| 3 | 40 | 40 |
| 4 | 15 | 40 |

Largest = 40

Smallest element using array

```c
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*****************************************************************************/
#include <stdio.h>

int main() {
    int arr[5], i, min;

    for(i = 0; i < 5; i++)
        scanf("%d", &arr[i]);

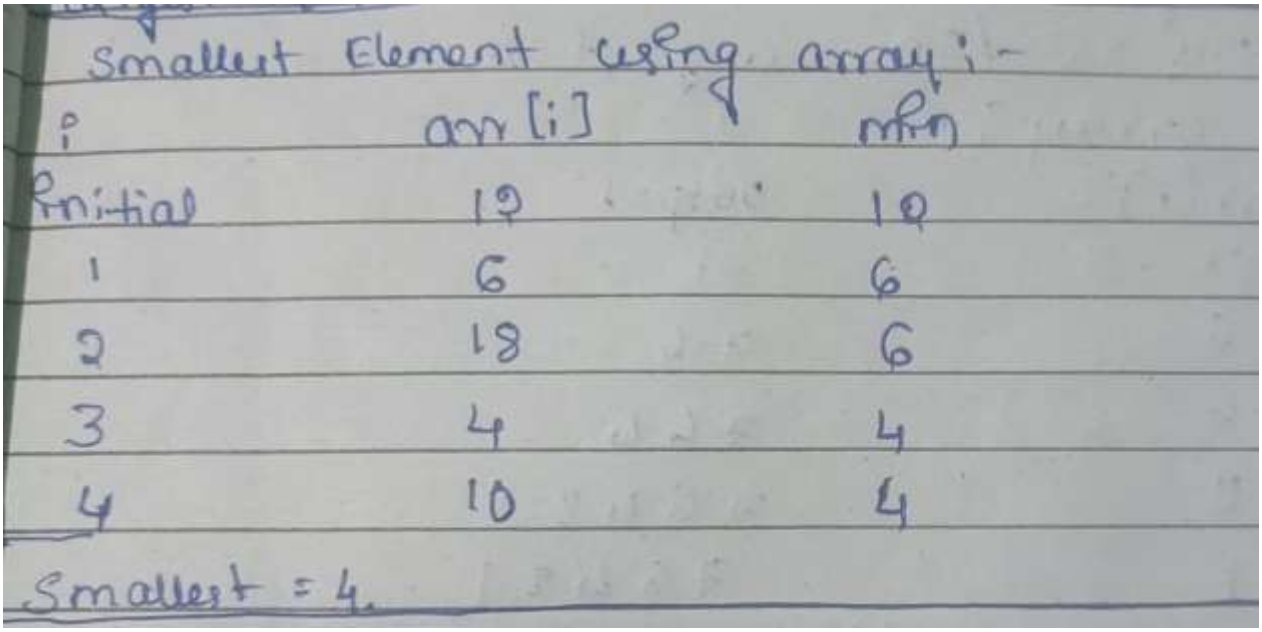
    min = arr[0];

    for(i = 1; i < 5; i++) {
        if(arr[i] < min)
            min = arr[i];
    }

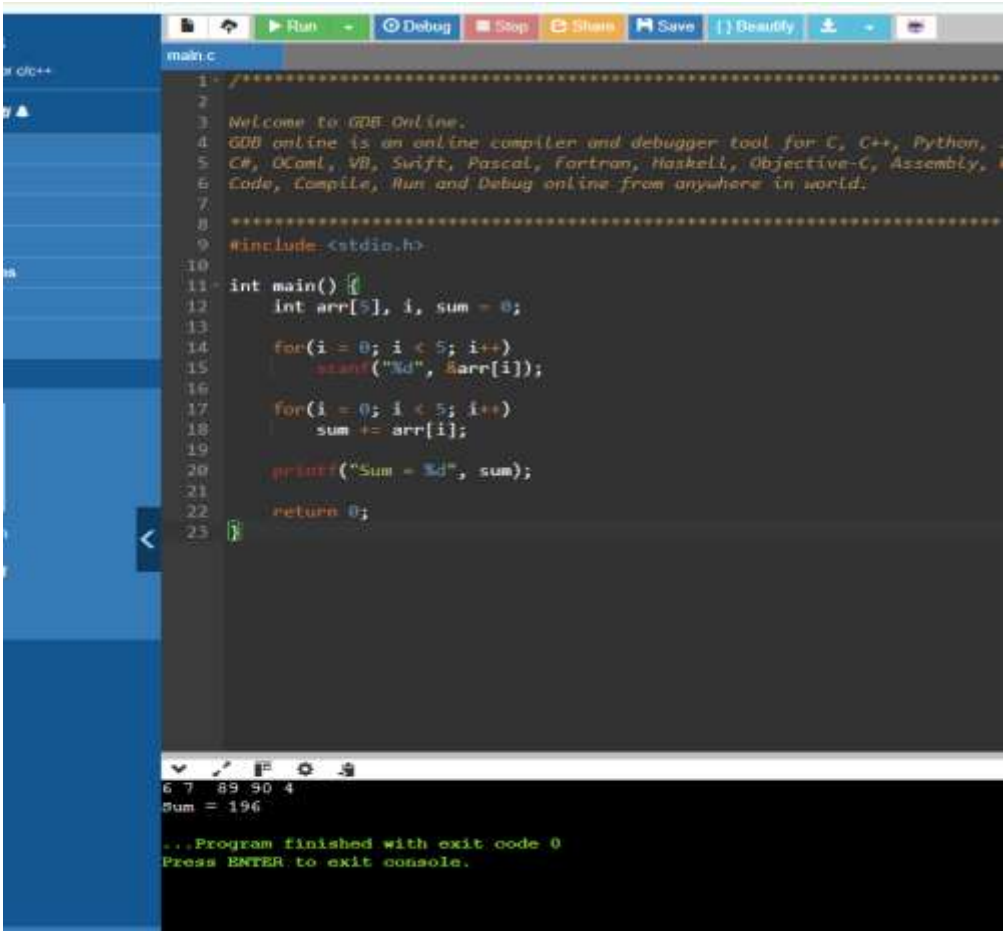
    printf("Smallest = %d", min);
    return 0;
}
```

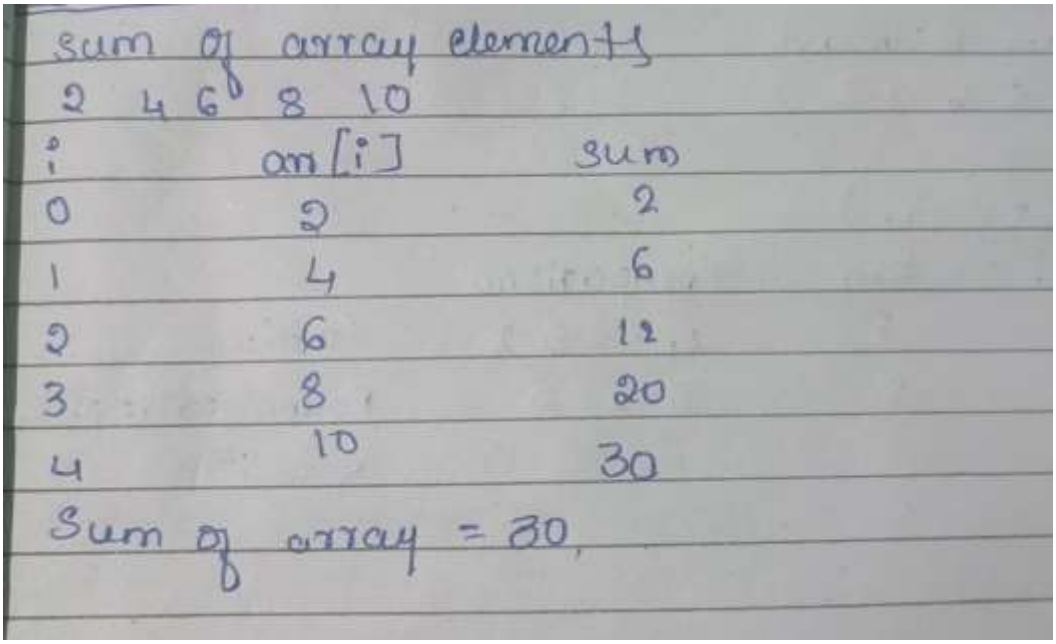
```
input
4 56 78 3 12
Smallest = 3

...Program finished with exit code 0
Press ENTER to exit console.
```

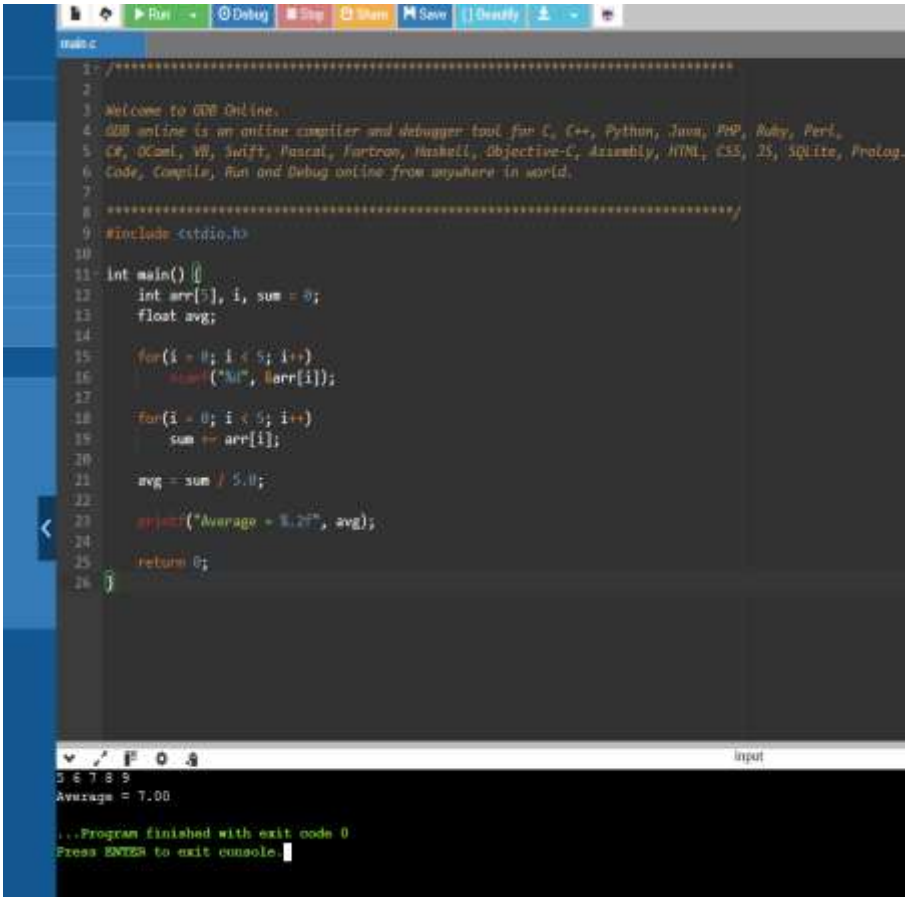


Sum of array elements



Average of array elements



```
/*****************************************************************************

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*****************************************************************************/
#include <stdio.h>

int main() {
    int arr[5], i, sum = 0;
    float avg;

    for(i = 0; i < 5; i++)
        scanf("%d", &arr[i]);

    for(i = 0; i < 5; i++)
        sum += arr[i];

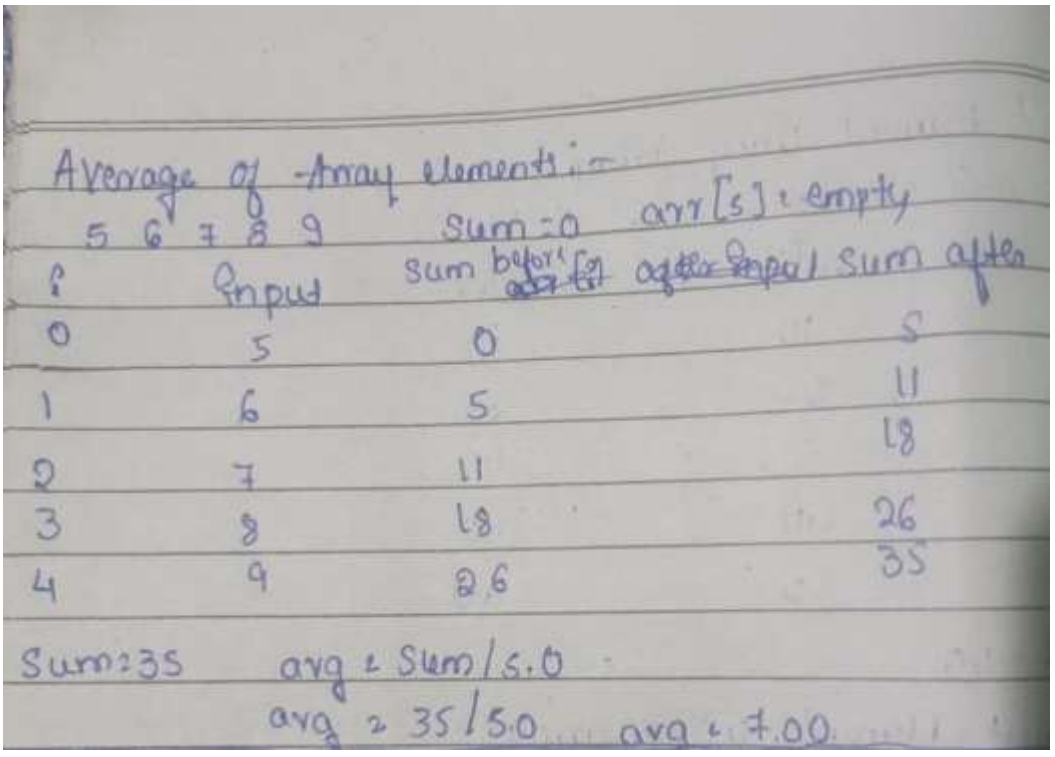
    avg = sum / 5.0;

    printf("Average = %.2f", avg);

    return 0;
}
```

```
Input
5 6 7 8 9
Average = 7.00

...Program finished with exit code 0
Press ENTER to exit console.
```



Average of Array elements:-

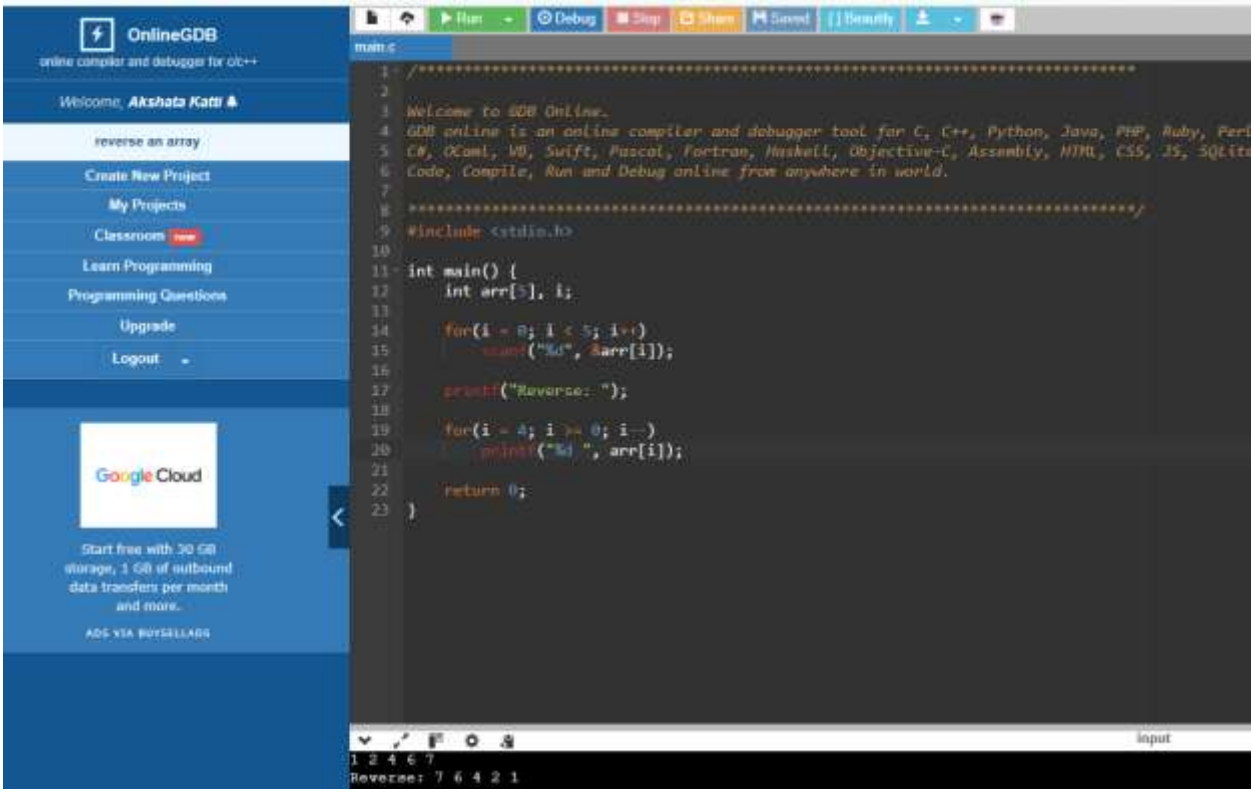
5 6 7 8 9 Sum=0 arr[5] : empty

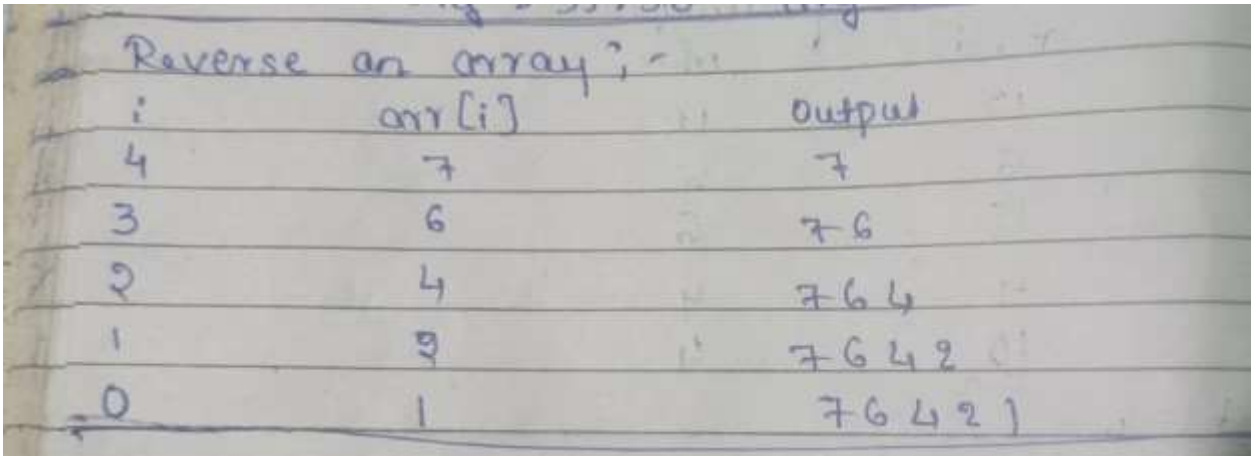
| i | Input | Sum before adding Input | Sum after |
|---|---|---|---|
| 0 | 5 | 0 | 5 |
| 1 | 6 | 5 | 11 |
| 2 | 7 | 11 | 18 |
| 3 | 8 | 18 | 26 |
| 4 | 9 | 26 | 35 |

Sum=35 avg = Sum/5.0

avg = 35/5.0 avg = 7.00

Reverse an array



Search an element

```c
/******************************************************************************

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*******************************************************************************/
#include <stdio.h>

int main() {
    int arr[5], i, key;

    for(i = 0; i < 5; i++)
        scanf("%d", &arr[i]);

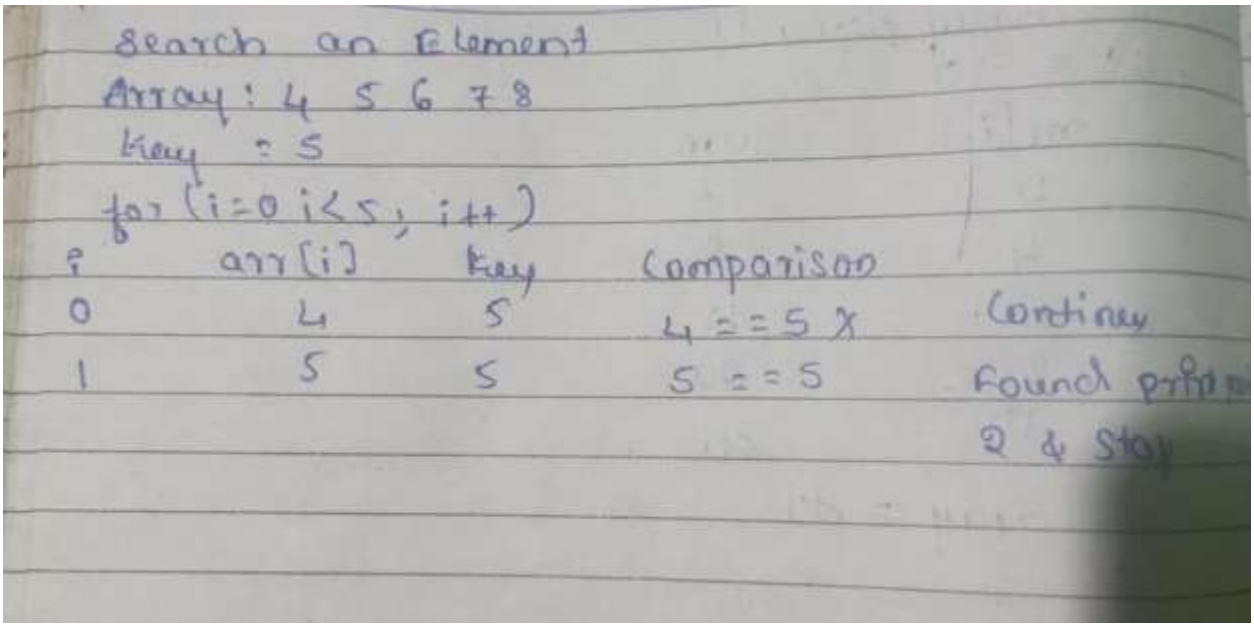
    scanf("%d", &key);

    for(i = 0; i < 5; i++) {
        if(arr[i] == key) {
            printf("Found at position %d", i + 1);
            return 0;
        }
    }

    printf("Not Found");

    return 0;
}
```

```
Input
4 5 6 7 8
5
Found at position 2
```



Count Even or odd elements

```c
/*****************************************************************************

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*****************************************************************************/
#include <stdio.h>

int main() {
    int arr[5], i, even = 0, odd = 0;

    for(i = 0; i < 5; i++)
        scanf("%d", &arr[i]);

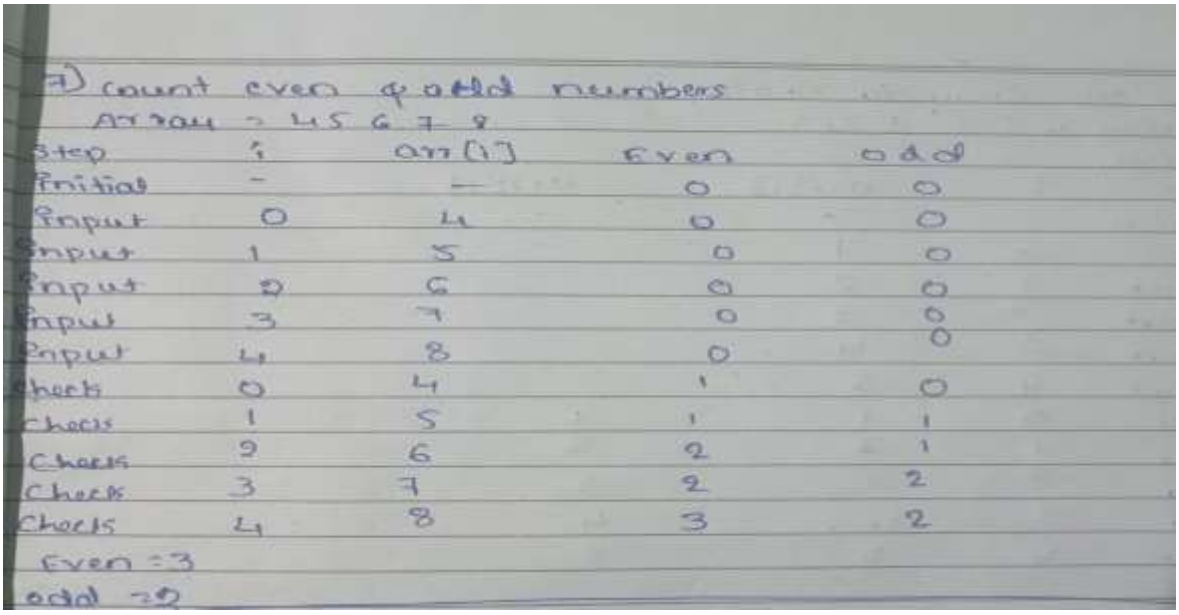
    for(i = 0; i < 5; i++) {
        if(arr[i] % 2 == 0)
            even++;
        else
            odd++;
    }

    printf("Even=%d\nOdd=%d", even, odd);

    return 0;
}
```

```
input
4 5 6 7 8
Even=3
Odd=2

...Program finished with exit code 0
Press ENTER to exit console.
```



Sort array ascending order

```c
/*****************************************************************************

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*****************************************************************************/
#include <stdio.h>

int main() {
    int arr[5], i, j, temp;

    for(i = 0; i < 5; i++)
        scanf("%d", &arr[i]);

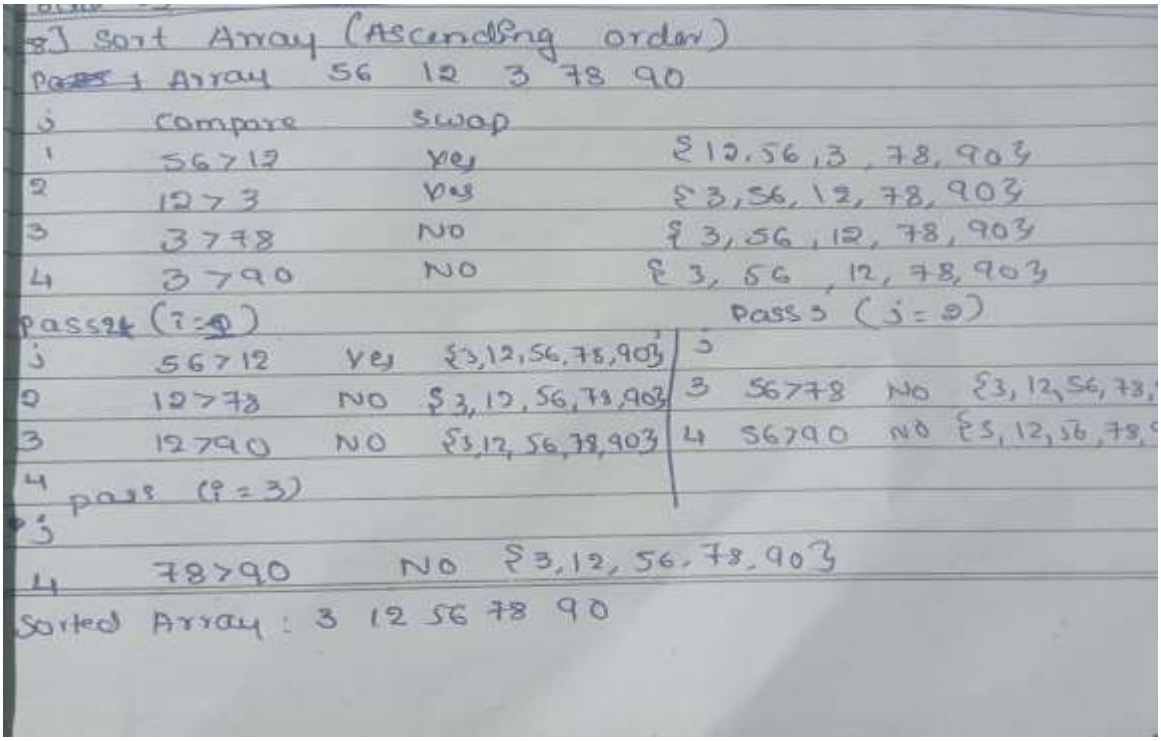
    for(i = 0; i < 4; i++) {
        for(j = i + 1; j < 5; j++) {
            if(arr[i] > arr[j]) {
                temp = arr[i];
                arr[i] = arr[j];
                arr[j] = temp;
            }
        }
    }

    printf("Sorted Array: ");

    for(i = 0; i < 5; i++)
        printf("%d ", arr[i]);

    return 0;
}
```

```
input
56 12 3 78 90
Sorted Array: 3 12 56 78 90
```



8] Sort Array (Ascending order)

Pass 1 Array 56 12 3 78 90

| j | Compare | Swap | |
|---|---|---|---|
| 1 | 56 > 12 | Yes | {12, 56, 3, 78, 90} |
| 2 | 12 > 3 | Yes | {3, 56, 12, 78, 90} |
| 3 | 3 > 78 | NO | {3, 56, 12, 78, 90} |
| 4 | 3 > 90 | NO | {3, 56, 12, 78, 90} |

Pass2 (i=1)

| j | | | |
|---|---|---|---|
| 2 | 56 > 12 | Yes | {3, 12, 56, 78, 90} |
| 3 | 12 > 78 | NO | {3, 12, 56, 78, 90} |
| 4 | 12 > 90 | NO | {3, 12, 56, 78, 90} |

Pass3 (i=2)

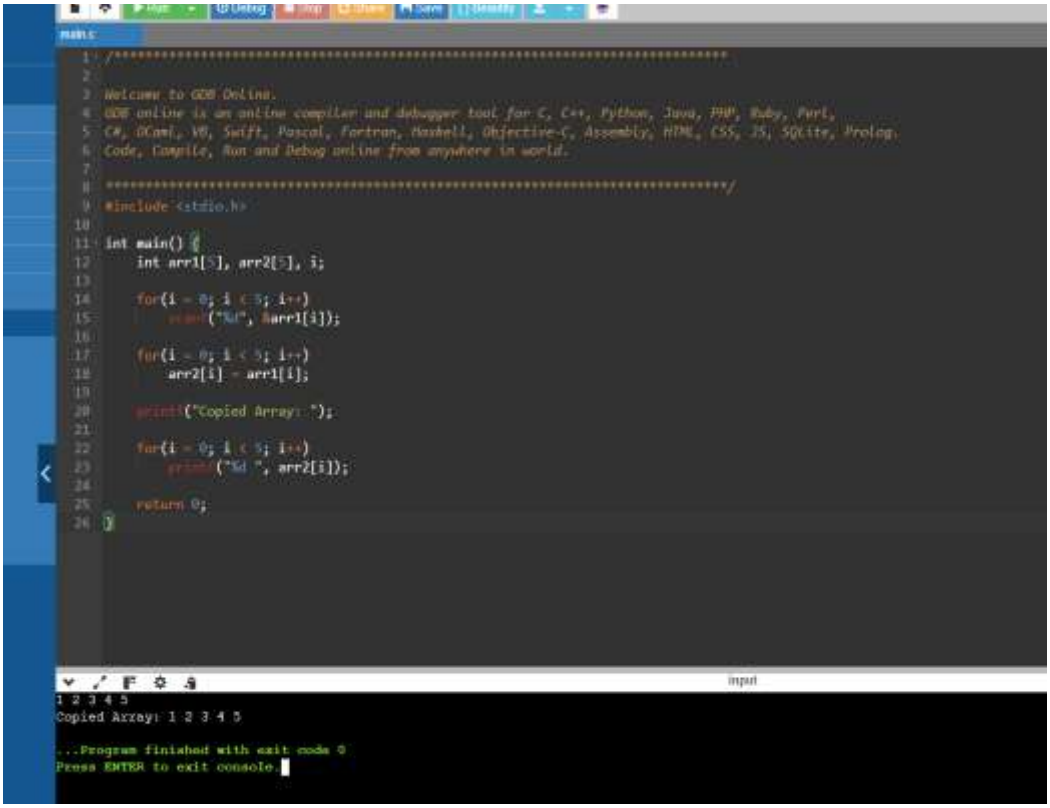
| j | | | |
|---|---|---|---|
| 4 | 78 > 90 | NO | {3, 12, 56, 78, 90} |

Pass 3 (i=2)

| j | | | |
|---|---|---|---|
| 3 | 56 > 78 | NO | {3, 12, 56, 78, |
| 4 | 56 > 90 | NO | {3, 12, 56, 78, |

Sorted Array : 3 12 56 78 90

Copy one array to another array



```c
/******************************************************************************

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*******************************************************************************/
#include <stdio.h>

int main() {
    int arr1[5], arr2[5], i;

    for(i = 0; i < 5; i++)
        scanf("%d", &arr1[i]);

    for(i = 0; i < 5; i++)
        arr2[i] = arr1[i];

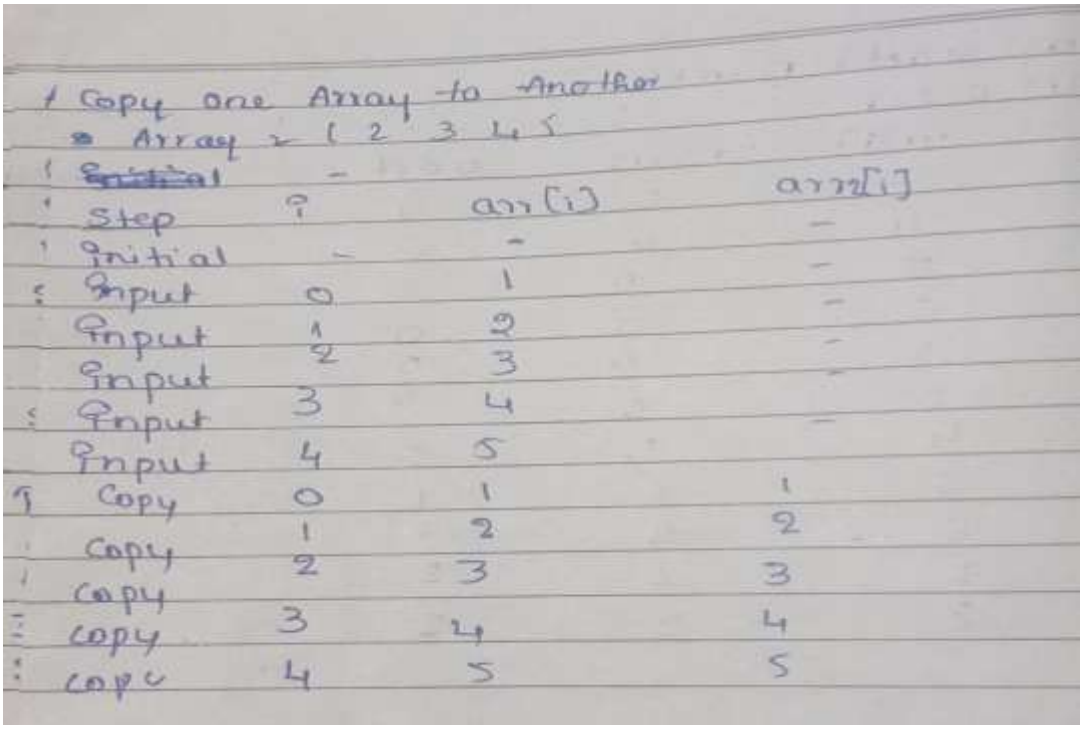
    printf("Copied Array: ");

    for(i = 0; i < 5; i++)
        printf("%d ", arr2[i]);

    return 0;
}
```

```
input
1 2 3 4 5
Copied Array: 1 2 3 4 5

...Program finished with exit code 0
Press ENTER to exit console.
```



1 Copy one Array to Another

Array = { 1 2 3 4 5 }

| Step | i | arr1[i] | arr2[i] |
|---|---|---|---|
| Initial | - | - | - |
| Input | 0 | 1 | - |
| Input | 1 | 2 | - |
| Input | 2 | 3 | - |
| Input | 3 | 4 | - |
| Input | 4 | 5 | - |
| Copy | 0 | 1 | 1 |
| Copy | 1 | 2 | 2 |
| Copy | 2 | 3 | 3 |
| Copy | 3 | 4 | 4 |
| Copy | 4 | 5 | 5 |